



Lab: Analyze AI Opportunities During the Product Concept Phase

Estimated duration: 20 minutes

Introduction

The purpose of a Product Concept is to identify a potential solution that addresses a defined problem or market need. In this lab, you will act as a Product Manager tasked with developing both a Product Concept and a Business Case. Using Google Gemini, you will explore how Generative AI can enhance the product development process and provide input to support your business case.

Learning objectives

By the end of this lab, you will be able to:

- Analyze how Generative AI can improve the product development process
- Use Generative AI to draft a business case aligned with the Balanced Scorecard framework

Prerequisites

- Navigate to <https://gemini.google.com/> using your web browser to access Google Gemini.

Scenario

Product

Company: ABC Homes Inc.

Industry: Household Cleaners

ABC Homes produces a variety of household cleaners in a highly competitive market. Customers expect products that are reasonably priced, environmentally safe, and effective.

Problem/Challenges

- Developing less toxic cleaners requires extensive research, testing, and formulation. ABC Homes currently relies on trial-and-error methods that are time-consuming and costly.
- A Product Manager believes Generative AI could accelerate and improve this process. They need to determine its value and present a business case to leadership for approval.

Current development process

1. **Product research:** Scientists study existing chemicals and their effects on health and the environment.
2. **Market research:** Marketing analyzes competitor products, their strengths/weaknesses, and lifecycle stages.
3. **Formulation:** Chemists experiment with ingredients to create safe, effective cleaners.
4. **Testing:** New products are developed, analyzed, and tested; refinements ensure safety and performance.
5. **Evaluation:** Based on test results, refinements are made, feasibility is evaluated, and business cases are developed.
6. **Production and distribution:** Approved products are mass-produced and distributed.

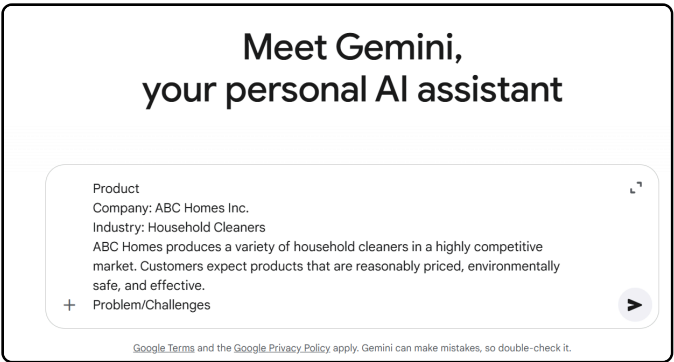
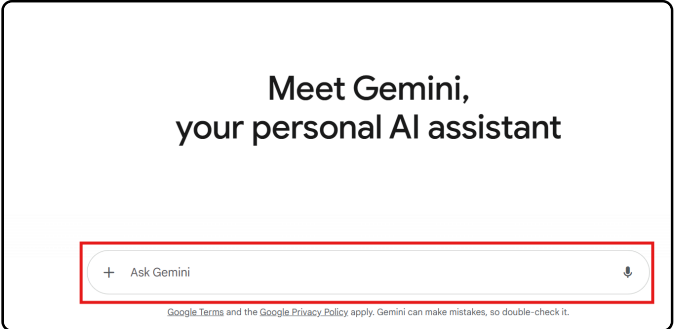
Tasks

Task 1: Identify AI opportunities in the current process

Your sponsor is not familiar with Generative AI and has asked you to provide the advantages of applying it to the current 6-step process. Steps:

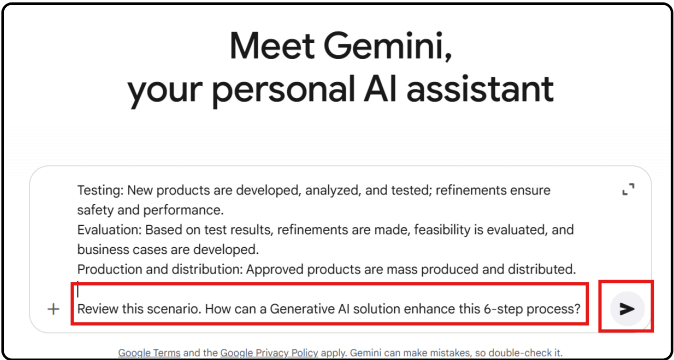
1. Open [Gemini](#) and copy the Product, Problem, and Challenges, and Current Development Process sections in the **Ask Gemini** field.

Product
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Industry: Household Cleaners
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Problem/Challenges
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• A Product Manager believes Generative AI could accelerate and improve this process. They need to determine its value and present a business case
Current development process
7. Product research: Scientists study existing chemicals and their effects on health and the environment.
8. Market research: Marketing analyzes competitor products, their strengths/weaknesses, and lifecycle stages.
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10. Testing: New products are developed, analyzed, and tested; refinements ensure safety and performance.
11. Evaluation: Based on test results, refinements are made, feasibility is evaluated, and business cases are developed.
12. Production and distribution: Approved products are mass-produced and distributed.



2. Copy and paste the following prompt below the product details and press enter on your keyboard or click the send button.

Review this scenario. How can a Generative AI solution enhance this 6-step process?



3. Google Gemini will provide you with an in-depth analysis. Save the output for later analysis and sharing.

Overview: Business case focus

A strong business case shows the impact of a solution in one or more of the following four areas (Balanced Scorecard framework):

- Finance: Impact on revenue, cost savings, or cost avoidance.
- Process: Improvements in efficiency or effectiveness.
- Employees: Benefits for staff (e.g., reduced workload, enhanced collaboration).

- Customers: Added value for end users (e.g., improved safety, quality, or satisfaction).

Task 2: Build the business case with generative AI

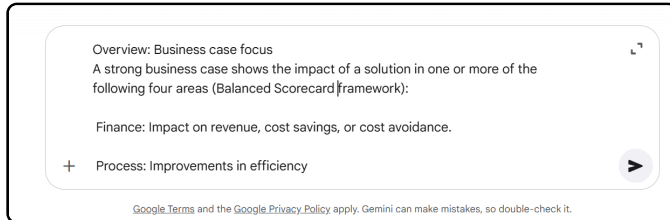
Now use Generative AI to connect the Product Concept to the Balanced Scorecard.

Steps:

1. Copy the four categories listed in the Overview section.

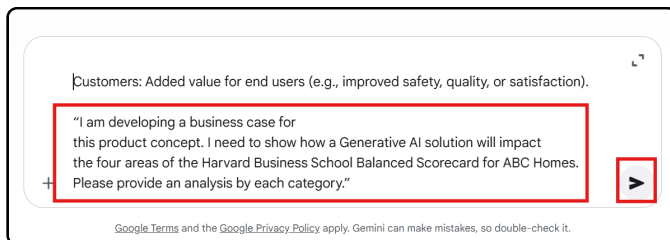
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2. Copy and paste the following prompt below the overview and press enter on your keyboard or click the send button.

I am developing a business case for this product concept. I need to show how a Generative AI solution will impact the four areas of the Harvard Business School Balanced Scorecard for ABC Homes.



3. Again, Google Gemini will provide you with a comprehensive analysis. Save the output for later analysis and sharing.

Author(s)

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